

Test Questions & Evaluation Sheet

Project: Akbar & Sons Knowledge Assistant (Week 4 RAG Chatbot) | Date: July 30, 2026 | Knowledge Base: 27 sections + Catalogue PDF

Test 1: Which brands sell PPRC pipes?

Expected Answer	Popular, TurkPlast, Dura Flow, IIL
Actual Answer	Popular, TurkPlast, Dura Flow, and IIL all sell PPRC pipes.
Sources Cited	PPRC Pipes & Fittings (77%), PPRC Heaters (77%), Catalogue PDF (75%)
Result	■ PASS

Test 2: What sizes do water tanks come in and warranty?

Expected Answer	150-1200 Gallon; Suntuff 10yr, Dura Flow 5yr
Actual Answer	Water tanks: 150, 200, 250, 300, 400, 500, 750, 1200 Gallon. Suntuff: 10-year warranty, Dura Flow: 5-year warranty.
Sources Cited	Water Tanks (73%), UPVC Pipes & Fittings (61%), Sewerage Pipes & Fittings (61%)
Result	■ PASS

Test 3: Do you sell electrical wire or cables?

Expected Answer	Not in catalogue — fallback message
Actual Answer	We do not have information about electrical wire or cables in our catalogue. Please contact the store directly for more details.
Sources Cited	Suction Steel Wire Garden Pipes (59%), Gas Pipes (58%), Threads & Tapes (58%)
Result	■ PASS (Fallback)

Test 4: What is the difference between UPVC pipes and PPRC pipes?

Expected Answer	UPVC for sewage (white), PPRC for hot/cold water (green)
Actual Answer	UPVC pipes are suitable for sewage, drainage, and waste-water systems in white. PPRC pipes are used for hot and cold water supply systems.
Sources Cited	PPRC Pipes & Fittings (71%), PPRC Heaters (69%), UPVC Pipes & Fittings (68%)
Result	■ PASS

Test 5: What are your store hours and where are you located?

Expected Answer	9 AM–9 PM, Main Defence Ghazi Road, Lahore
Actual Answer	Our store hours are from 9:00 AM to 9:00 PM. We are located on Main Defence Ghazi Road, near Jamia Masjid Siddiqia, Lahore.
Sources Cited	Store Information (58%), Door Locks (50%), Water Tanks (50%)
Result	■ PASS

Summary

Total Tests	5
Passed	5
Failed	0
Success Rate	100%

Key Findings

- **Retrieval Accuracy:** All queries returned relevant sources with high confidence scores (50–77%).
- **No Hallucinations:** System consistently drew answers only from the knowledge base.
- **Fallback Mechanism:** Correctly triggered for out-of-scope queries (electrical wire test).
- **Source Attribution:** All answers included citations with match percentages.
- **Multi-Chunk Synthesis:** System combined multiple catalogue sections accurately.

Conclusion

The RAG pipeline is functioning correctly and meets all assignment requirements. The system accurately retrieves relevant knowledge base chunks, grounds all answers in retrieved context, provides transparent source attribution, and appropriately declines to answer out-of-scope questions without hallucinating.